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# — MNIST Classification – MLP (Keras style from your Exp04) —

import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.utils import to_categorical
from sklearn.metrics import classification_report, confusion_matrix
import pandas as pd

# 1. Load & preprocess (exactly like your Exp04)
(X_train, y_train), (X_test, y_test) = mnist.load_data()

X_train_flat = X_train.reshape(60000, 784)
X_test_flat = X_test.reshape(-1, 28*28)

X_train_sc = X_train_flat.astype('float32') / 255.0
X_test_sc = X_test_flat.astype('float32') / 255.0

y_train_cat = to_categorical(y_train)
y_test_cat = to_categorical(y_test)

# 2. Build MLP (matches your Exp04 Cell 36)
model = Sequential()
model.add(Dense(512, input_dim=784, activation='relu'))
model.add(Dense(256, activation='relu'))
model.add(Dense(128, activation='relu'))
model.add(Dense(32, activation='relu'))
model.add(Dense(10, activation='softmax'))

model.summary()

# 3. Compile & Train
model.compile(loss='categorical_crossentropy',
              optimizer='rmsprop',
              metrics=['accuracy'])

h = model.fit(X_train_sc, y_train_cat,
              batch_size=128,
              epochs=10,
              verbose=1,
              validation_split=0.1)

# 4. Evaluate
train_acc = model.evaluate(X_train_sc, y_train_cat, verbose=0)[1]
test_acc = model.evaluate(X_test_sc, y_test_cat, verbose=0)[1]
print(f"Train Accuracy: {train_acc:.4f}")
print(f"Test Accuracy: {test_acc:.4f}")

# 5. Plot accuracy
plt.plot(h.history['accuracy'])
plt.plot(h.history['val_accuracy'])
plt.legend(['Training', 'Validation'])
plt.title('Accuracy')
plt.xlabel('Epochs')
plt.show()

# 6. Classification report
y_pred = np.argmax(model.predict(X_test_sc), axis=1)
print(classification_report(y_test, y_pred))

# 7. Bar chart for accuracy comparison
labels = ['Train Accuracy', 'Test Accuracy']
values = [train_acc, test_acc]

plt.figure()
plt.bar(labels, values)
plt.title('Train vs Test Accuracy')
plt.ylabel('Accuracy')
plt.ylim(0, 1)
plt.show()

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